AGE OF AVAX × TEAM1

Research and development · Avalanche ecosystem

Age of Avax

An RTS with a verifiable economy on its own Avalanche blockchain



TECHNICAL AND ECONOMIC WHITEPAPER · v1.4

Version	1.4 (supersedes 1.3)
Date	October 2026
Authors	Age of Avax team, with Team1
Network	Age L1 · Avalanche (Fuji: chain ID 777011)
Game	https://ageofavax.app
Code	https://github.com/MarxMad/Age-of-AVAX

This document describes a design under development. It is not an offer of securities nor investment advice. GLORY is an in-game currency, not an investment instrument.

Abstract

Age of Avax is a 3D real-time strategy (RTS) game with the classic *Age of Empires* loop, whose economy runs on its own blockchain: the *Age L1*, built with Subnet-EVM on Avalanche. Matches run in the browser; only what carries value goes on-chain: asset ownership, rewards and verifiable results.

The simulation is deterministic: a seed and the command log reproduce the exact match on any machine and browser. A verifier re-simulates each victory and signs it (EIP-712), and a relayer reports it to *MatchRegistry*, paying the gas. After a dispute window, the *RewardsPool* pays the winner. Players start for free as guests and claim when they connect a wallet, without needing AVAX or AGE.

The game currency, GLORY, has a **fixed supply of 100 million**, minted once. Rewards come from a 40 M pool with a daily budget that halves every year, per-player caps, diminishing returns and anti-bot defenses both on-chain and on the server. The business model charges in AVAX, USDC or card (sponsorships, season pass, cosmetics and tournaments), not through token sales.

A working prototype validates the design: a 3D game with seven ages, from Ancient to Futuristic, that change the architecture; an AI with four difficulty levels; 1 vs 1 lockstep multiplayer; nine contracts with 19 tests plus 77 simulation tests, and the full play, verify, report and claim loop running end to end.

Keywords: Avalanche, L1, Subnet-EVM, RTS, deterministic simulation, lockstep, tokenomics, fixed supply, anti-farming, Team1.

Contents

1. Introduction 1

1.1 Context 1

1.2 Problem and response 1

1.3 Contributions 1

2. Game design 2

2.1 Core loop 2

2.2 Ages 2

2.3 Buildings 4

2.4 Units 5

2.5 Civilizations and Team1 identity 7

2.6 In-match sinks 7

2.7 Game modes 8

3. Technical architecture 9

3.1 Overview 9

3.2 Deterministic simulation 10

3.3 1 vs 1 lockstep multiplayer 10

3.4 3D client and performance 10

3.5 Game server 10

4. The Age L1 on Avalanche 11

4.1 Why an own L1 11

4.2 Parameters 11

4.3 Operating cost 11

5. How it works on-chain 12

5.1 What goes on-chain 12

5.2 Player journey 12

5.3 Verifiable matches 12

6. Tokenomics 14

6.1 GLORY: utility . . . 14
6.2 Fixed supply and allocation . . . 14
6.3 Reward issuance . . . 15
6.4 Sinks . . . 16
6.5 Other tokens . . . 16

7. Anti-farming and economic security . . . 17
7.1 On-chain defenses . . . 17
7.2 Server-side defenses . . . 17
7.3 Worst case . . . 18

8. Assets, marketplace and pools . . . 19
8.1 ERC-1155 assets . . . 19
8.2 Marketplace . . . 19
8.3 Resource pools . . . 19
8.4 Tokenized resources: FOOD, WOOD, GOLD and STONE . . . 20

9. Business model . . . 21
9.1 Principle . . . 21
9.2 Revenue streams . . . 21
9.3 Costs and scalability . . . 21
9.4 Illustrative first-year scenario . . . 22

10. Roadmap and coming soon . . . 23
10.1 Upcoming buildings . . . 23
10.2 Upcoming ages and civilizations . . . 23

11. Risks and regulatory considerations . . . 24

12. Conclusions . . . 25

References . . . 26

Appendix A. Contracts . . . 27

List of figures and tables

Table 1. Common failures and the Age of Avax response. . . . 1

Figure 1. Town center (Primary Network) by age: sandstone and timber, marble with towers, cannons, brick with chimneys, glass with a spire and LED outline, and the white structure of the Futuristic Age. Bottom right: robots, plasma tanks and helicopters. . . . 2

Table 2. Ages: advancement cost and what each one unlocks. . . . 3

Figure 2. A selection of buildings (captures from the 3D engine). . . . 4

Table 3. Building catalog (names of the Avalanche Pioneers civilization). . . . 4

Table 4. Unit catalog. . . . 6

Table 5. Civilizations. . . . 7

Figure 3. Alpine Shelters in the colors of the Team1 chapters; the band at the foot of the wall shows the owner. . . . 7

Figure 4. Architecture: client, game server and Age L1. . . . 9

Table 6. Age L1 parameters on Fuji. . . . 11

Table 7. What is recorded and where. . . . 12

Figure 5. Verification, reporting, dispute and automatic claim. . . . 13

Table 8. GLORY allocation. . . . 14

Figure 6. Allocation of the fixed GLORY supply. . . . 14

Table 9. RewardsPool emission ceiling. . . . 15

Figure 7. Annual emission ceiling and cumulative. It is a maximum: fewer players means less issuance. . . . 16

Table 10. GLORY sinks. . . . 16

Table 11. System tokens. . . . 16

Table 12. Hard limits in contracts. . . . 17

Table 13. Verifier filters. . . . 17

Table 14. Rarities and reference prices. . . . 19

Table 15. x·y=k pools against GLORY (0.3% fee). . . . 19

Table 16. Life cycle of a resource token. . . . 20

Table 17. Settlement by result. . . . 20

Table 18. Revenue streams in order of activation. . . . 21

Table 19. Cost structure. . . . 21

Table 20. Scenario with explicit assumptions (not a projection). . . . 22

Figure 8. Phased roadmap with its gates. . . . 23

Table 21. Risks and mitigations. 24

Table 22. Age of Avax contracts. 27

Chapter 1

Introduction

1.1 Context

Tokenized games promised true ownership of digital assets, yet the first generation failed mostly because of economic design: unlimited rewards, inflationary currencies and value sustained only by the arrival of new players. Traditional games, meanwhile, lock items on private servers, so results cannot be verified and earnings cannot be transferred.

1.2 Problem and response

Table 1. Common failures and the Age of Avax response.

Common failure	Consequence	Age of Avax response
Unlimited issuance ("play-to-earn")	The currency inflates and collapses	Fixed 100 M GLORY; rewards from a pool with yearly halving
Bots and account farms	Rewards extracted without playing	Global daily budget, per-player caps, single-use seeds, real-time floor
Unverifiable results	Cheating and undue payouts	Deterministic simulation, re-simulation and on-chain disputes
Onboarding friction (wallet, gas)	Few new players	Free guest play; gasless claims; sponsored gas
Shared chain	Expensive, unpredictable gas	Own L1: gas in AGE, 1 s blocks

1.3 Contributions

- An architecture that separates real-time play (browser) from economic facts (chain), with results verifiable by re-simulation.
- 1 vs 1 multiplayer through deterministic lockstep, with bit-exact math across browsers.
- Fixed-supply tokenomics with issuance bounded by construction and layered anti-farming defenses.
- A "play for free, connect to earn" journey with no gas friction.
- Ecosystem identity: a town center shaped like the Avalanche logo, an academy shaped like the Team1 t1 mark, and houses in the colors of the Team1 chapters.

Chapter 2

Game design

2.1 Core loop

Each player starts with a town center, four villagers and a scout. They gather food, wood, gold and stone; build; train an army; and advance through the ages. Destroying the enemy town center wins. In-match resources are not tokens: they reset every game, unless they are bought or settled through the vault (Chapter 8).

Names below are English translations of the in-game names (the game UI is currently in Spanish).

2.2 Ages

As in AoE, every age changes how buildings look and unlocks units, buildings and upgrades. The town center keeps the Avalanche logo silhouette but is built from each era's materials (Figure 1).

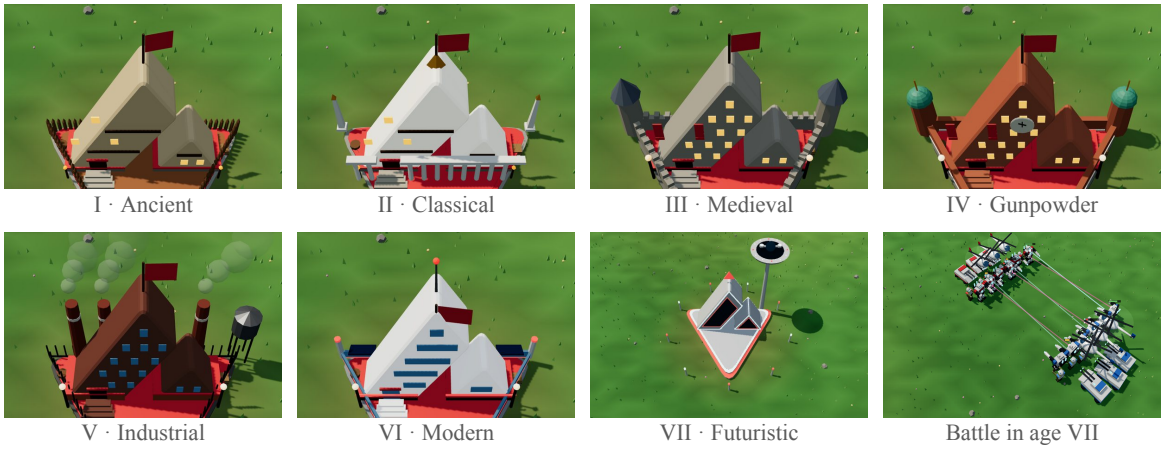


Figure 1. Town center (Primary Network) by age: sandstone and timber, marble with towers, cannons, brick with chimneys, glass with a spire and LED outline, and the white structure of the Futuristic Age. Bottom right: robots, plasma tanks and helicopters.

The Futuristic Age (VII) rebuilds every structure in white ceramic, graphite and dark glass with neon edges, and brings combat robots, helicopters that fly over water, forests and walls, and hovering plasma tanks. They all fire lasers.

Table 2. Ages: advancement cost and what each one unlocks.

Age	Advance	Buildings	Units	Upgrades
I · Ancient	—	Primary Network, Alpine Shelter, Node Granary, Firewood Sawmill, AVAX Mine, Staking Farm, Consensus Barracks, Ice Palisade, DeFi Market, Teleporter Port, Glacier Gate	Validator, Sentinel, Coreth Fisher, ICM Ferry	Double-bit axe
II · Classical	400 food · 100 wood	Snowball Range, Snowflake Stable, HyperSDK Forge, Validator Tower, Glacier Wall	Snowball Marksman, Snowflake Scout, DeFi Caravan, Interchain Freighter, Snowman Sage, Avalanche Galley	Wheelbarrow, Plough, Mining pick, Forging
III · Medieval	700 food · 300 gold	Subnet Workshop, L1 Fortress	Avalanche Knight, Glacier Catapult	Plate armor, Horse collar, Bow saw, Warehouses, Masonry
IV · Gunpowder	900 food · 600 gold · 200 stone	Team1 Academy, Temple of Consensus	C-Chain Musketeer, Subnet Cannon	Ballistics, Deep mining, Blast furnace
V · Industrial	1,200 food · 900 gold · 400 stone	Avalanche9000 Factory	Warp Rifleman	Chemistry, Crop rotation, Armor plating, Assembly line
VI · Modern	1,500 food · 1,200 gold · 600 stone	—	Avalanche9000 Tank	Logistics, Precision guidance
VII · Futuristic	2,000 food · 1,600 gold · 800 stone	—	Validator Mech, Interchain Hawk, Etna Plasma Tank	—

2.3 Buildings

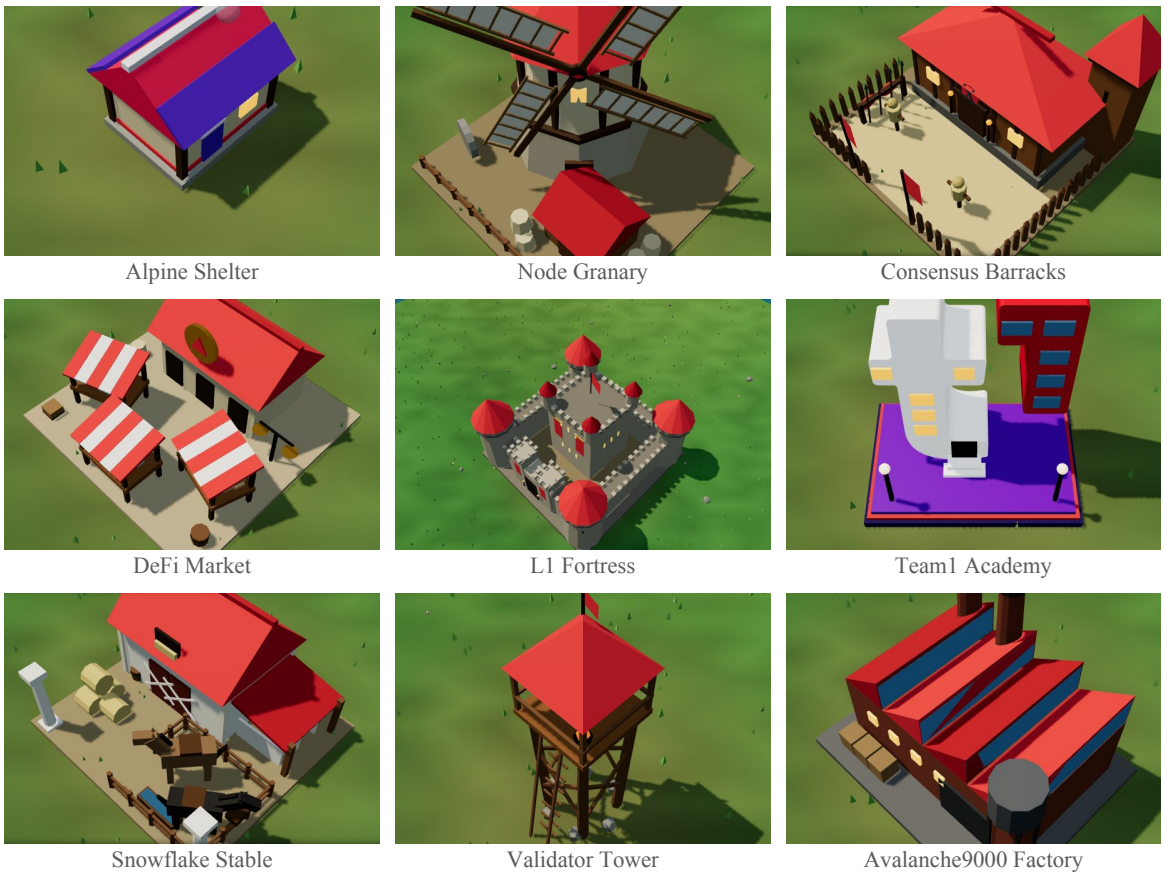


Figure 2. A selection of buildings (captures from the 3D engine).

Table 3. Building catalog (names of the Avalanche Pioneers civilization).

Building	Age	Cost	Role
Primary Network Town Center	I	275 wood · 100 stone	Trains Validators, receives every resource, researches the ages
Alpine Shelter House	I	30 wood	+5 population
Node Granary Mill	I	100 wood	Food drop-off; farming upgrades
Firewood Sawmill Lumber Camp	I	100 wood	Wood drop-off; logging upgrades
AVAX Mine Mining Camp	I	100 wood	Gold and stone drop-off; mining upgrades
Staking Farm Farm	I	60 wood	Renewable food
Consensus Barracks Barracks	I	100 wood · 40 stone	Infantry: Sentinel, Musketeer, Rifleman
Snowball Range Archery Range	II	175 wood	Ranged marksmen

Building	Age	Cost	Role
Snowflake Stable Stable	II	175 wood	Cavalry
HyperSDK Forge Blacksmith	II	150 wood	Attack and armor
Validator Tower Watch Tower	II	50 wood · 125 stone	Automatic defense
Subnet Workshop Siege Workshop	III	200 wood · 50 gold	Siege: catapults and cannons
Team1 Academy University	IV	200 wood · 100 gold · 100 stone	Ballistics, chemistry, guidance (Team1 t1 shape)
Avalanche9000 Factory Factory	V	300 wood · 200 gold · 200 stone	Cannons and tanks; assembly line
L1 Fortress Castle	III	650 stone	Fortress: shoots, houses 20 population, trains elite units
Ice Palisade Palisade	I	3 wood	Cheap log wall
DeFi Market Market	I	175 wood	Buys and sells against the L1 pools
Teleporter Port Dock	I	150 wood	On the shore: ships and fish drop-off
Temple of Consensus Wonder	IV	1,000 wood · 1,000 gold · 1,000 stone	Alternative victory: stand 5 minutes once finished
Glacier Gate Gate	I	40 wood	Inside a wall: your units pass, enemies do not
Glacier Wall Stone Wall	II	5 stone	Sturdy stone wall

Farms change with the age (wood and scarecrows, stone and hay, irrigation and tanks, neon and drones) and each one grows wheat, corn or pumpkins. A neutral **Grand Bazaar** sits at the center of every map as the destination for trade carts: each trip pays gold according to distance.

2.4 Units

Table 4. Unit catalog.

Unit	Age	Cost	HP	Attack	Range
Validator Villager	I	50 food	40	3	melee
Sentinel Militia	I	60 food · 20 gold	90	10	melee
Snowball Marksman Archer	II	40 wood · 40 gold	60	7	3.8 tiles
Snowflake Scout Scout	II	80 food	100	6	melee
Avalanche Knight Knight	III	70 food · 80 gold	180	14	melee
Glacier Catapult Catapult	III	160 wood · 120 gold	120	30	6.0 tiles
C-Chain Musketeer Musketeer	IV	60 food · 60 gold	100	18	4.2 tiles
Subnet Cannon Cannon	IV	150 wood · 200 gold	160	45	7.0 tiles
Warp Rifleman Rifleman	V	70 food · 70 gold	120	22	5.0 tiles
Avalanche9000 Tank Tank	VI	100 wood · 250 gold · 50 stone	450	50	5.5 tiles
Validator Mech Combat Robot	VII	90 food · 140 gold	260	30	4.8 tiles
Interchain Hawk Attack Helicopter	VII	120 wood · 260 gold	230	26	5.2 tiles
Etna Plasma Tank Plasma Tank	VII	120 wood · 320 gold · 80 stone	620	70	6.2 tiles
DeFi Caravan Trade Cart	II	100 food · 50 gold	70	0	—
Coreth Fisher Fishing Boat	I	75 wood	60	0	—
ICM Ferry Transport Ship	I	125 wood	160	0	—
Interchain Freighter Merchant Ship	II	100 wood · 50 gold	120	0	—
Snowman Sage Monk	II	100 gold	45	0	—
Avalanche Galley War Galley	II	140 wood · 50 gold	200	9	4.8 tiles

2.5 Civilizations and Team1 identity

The base civilization, **Avalanche Pioneers**, is free and rebrands everything with the ecosystem (Validator, Primary Network, AVAX Mine, Avalanche9000 Tank) without changing the mechanics. The other five are NFTs with moderate bonuses. Houses take the colors of the Team1 chapters and the Academy is shaped like the t1 mark (Figure 3).

Table 5. Civilizations.

Civilization	Bonus	How to get it
Avalanche Pioneers	Avalanche identity, no bonuses	Free
Glacier Norse	+15% woodcutting, villagers +10% speed	NFT 201
Subnet Empire	+15% gold and stone	NFT 202
C-Chain Guardians	Buildings +25% HP, defenses +2	NFT 203
P-Chain Riders	Cavalry +15% attack, +10% speed	NFT 204
X-Chain Archers	+30 ranged range	NFT 205



Figure 3. Alpine Shelters in the colors of the Team1 chapters; the band at the foot of the wall shows the owner.

2.6 In-match sinks

- **Tiered upgrades:** fourteen technologies across ages III to VI, each tier more expensive.
- **Paid repairs:** a full repair costs 50% of the building price.
- **Storage cap:** 800 + 400 per age + 150 per drop-off; excess decays 4% per minute.

2.7 Game modes

- **Versus AI:** entry mode, with four levels: Easy, Normal, Hard and Expert. Difficulty changes the AI income, how fast it advances ages, the size of its army and how often it attacks. From Normal up it pays a base of 500 GLORY per verified victory, plus the sale of any leftover resources; Easy is practice and pays nothing (the verifier rejects it).
- **1 vs 1 rooms:** 5-letter code or invite link; pays a base of 1,000 GLORY when both players connect a wallet.
- **Tournaments (Phase 2):** escrowed entry, sponsored prize pool, automatic payouts.

Maps: Mainland, Rivers, Islands, Coast, Desert, Mountains and Snow, 108 × 108 tiles, generated deterministically and symmetrically from the seed; everything around each base is mirrored. Gold and stone come in veins of several adjacent pieces, as in Age of Empires II: each base has two gold veins (3,600) and one stone vein (1,400) that several villagers can work at once.

Chapter 3

Technical architecture

3.1 Overview

The browser runs the simulation and rendering; the game server verifies, reports and coordinates 1 vs 1; the Age L1 stores ownership, rewards and results (Figure 4).

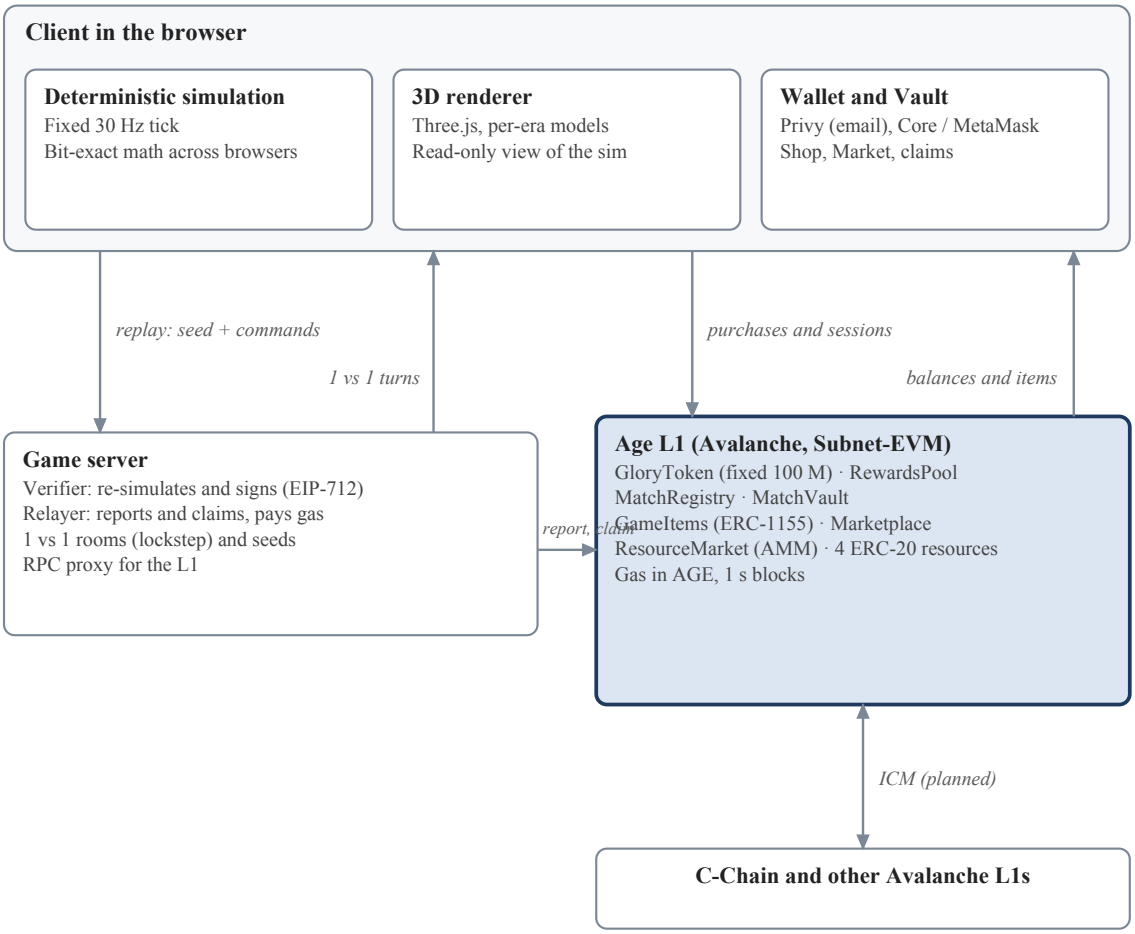


Figure 4. Architecture: client, game server and Age L1.

3.2 Deterministic simulation

The simulation advances in fixed 30 Hz ticks with a seeded pseudo-random generator. Let s be the seed, C the tick-stamped command sequence and f the one-tick transition; the state after T ticks is:

$$E_T = f^T(E_0(s), C) \quad (1)$$

Math.sin, *Math.cos* and *Math.hypot* may differ in the last bit across JavaScript engines, which is enough to desync. The simulation therefore uses its own library built only on addition, subtraction, multiplication, division and square root, which IEEE-754 defines exactly (error versus the native functions below 10^{-11}).

3.3 1 vs 1 lockstep multiplayer

No state is transmitted, only commands. A turn lasts 3 ticks (100 ms); what is ordered on turn T executes on turn $T + 3$ in both browsers, in the same order. The server filters and relays commands, keeps the log and compares state hashes every 5 s; at the end it re-simulates the log to certify the winner. Leaving, or 15 s without turns, counts as a loss.

3.4 3D client and performance

The client uses Three.js. Static parts of each model are merged into one mesh per material, which cuts draw calls from ~1,800 to ~590 per frame. Frame rate is capped at 60 fps, resolution at 1.5x on Retina screens, and shadows refresh every other frame. A power-saving mode drops to 30 fps.

3.5 Game server

- **Verifier:** re-simulates with the same client code and signs results and settlements (EIP-712).
- **Relayer:** sends report and claim, paying gas in AGE; players sign no transaction to get paid.
- **Rooms:** lockstep WebSocket; wallets prove ownership by signing a message containing the room code.
- **RPC proxy:** a single public origin; the L1 node is not exposed.

Chapter 4

The Age L1 on Avalanche

4.1 Why an own L1

A dedicated chain gives control over gas, performance and rules, without competing for blockspace with other applications. Gas is paid in AGE, which the protocol mints with NativeMinter and sponsors: players never buy it.

4.2 Parameters

Table 6. Age L1 parameters on Fuji.

Parameter	Value
Virtual machine	Subnet-EVM (Solidity, EVM wallets and tooling)
Chain ID	777011 (Fuji); 777010 on the local network
Gas token	AGE (18 decimals), sponsored by the protocol
Blocks	1 s target; 15,000,000 gas limit
Minimum base fee	1 gwei of AGE
Validation	Proof of Authority, 1 initial validator; Validator Manager v2
Precompiles	NativeMinter, FeeManager, Warp (ICM)
Admin	Dedicated deployer wallet; multisig on mainnet

4.3 Operating cost

Under ACP-77, each validator pays a continuous fee to the P-Chain (on the order of 1.33 AVAX per month) instead of locking 2,000 AVAX. Creating the L1 costs hundredths of an AVAX. The marginal cost per player is close to zero: matches run in the browser and gas is our own AGE.

Chapter 5

How it works on-chain

5.1 What goes on-chain

Table 7. What is recorded and where.

Data	Where it lives
Every click and every tick	Browser (plus the room server log in 1 vs 1)
Replay fingerprint (logHash) and final state	MatchRegistry
Winner, loser, seed and duration	MatchRegistry
Win/loss record	MatchRegistry
Rewards	RewardsPool → wallet
Resources bought and sold	MatchVault + ResourceMarket
Item ownership	GameItems (ERC-1155)

5.2 Player journey

- **Guest:** plays for free with a single-use seed issued by the server; a victory is stored for 24 h in the browser.
- **Signs in** with an email (Privy embedded wallet, no extension or seed phrase) or with a wallet (Core or MetaMask): receives 0.2 AGE of sponsored gas, topped up when it runs out, and claims pending victories.
- **Each new victory:** "Claim GLORY". The server re-simulates, reports, and claims automatically when the dispute window closes. On the same screen the player can settle resources: sell them for GLORY or keep them as tokens.
- **Asset owner:** NFT civilizations, army and blueprints, an in-game Market with real swaps, and rewarded 1 vs 1.

5.3 Verifiable matches

No reward is paid on the client's word (Figure 6). The signed result stays pending during a dispute window; anyone can re-simulate the replay from the anchored data and dispute it.

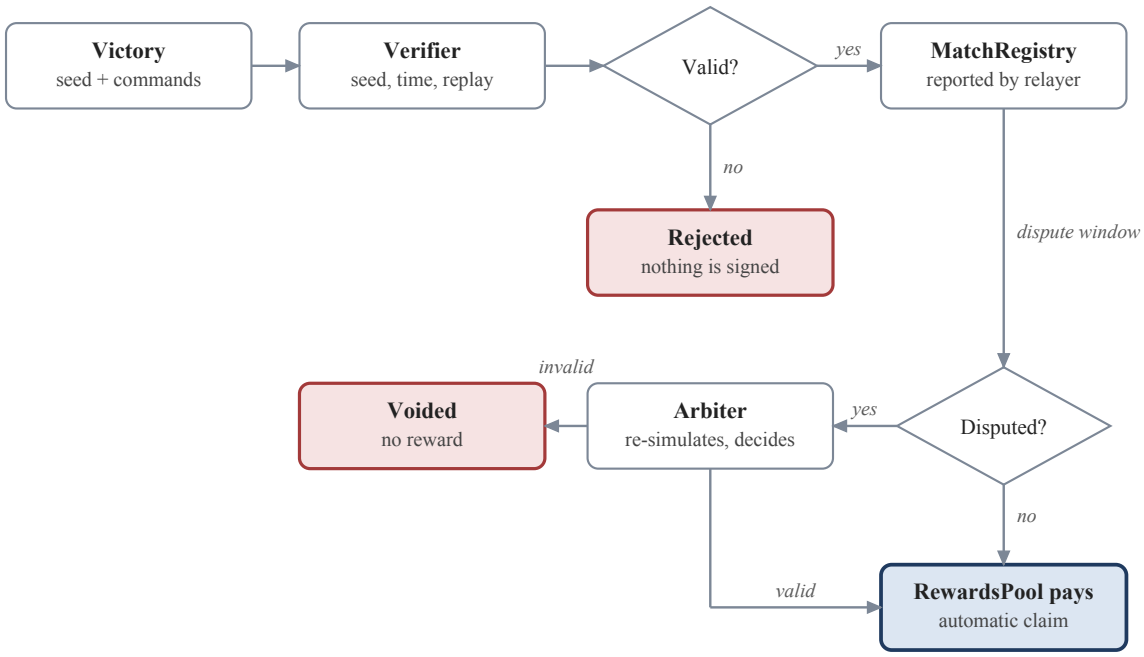


Figure 5. Verification, reporting, dispute and automatic claim.

Chapter 6

Tokenomics

6.1 GLORY: utility

GLORY is the game currency. It is earned by playing and spent in the Vault (skins, civilizations, army and blueprints), on the marketplace and on resources in the in-game Market. It is not sold as an investment and its value is not promised.

6.2 Fixed supply and allocation

Total supply is **100,000,000 GLORY**, all minted at deployment (*ERC20Capped*). The only minting permission is revoked at the end of deployment, so nobody can ever issue more: the script asserts *totalSupply* = *MAX_SUPPLY* before finishing.

Table 8. GLORY allocation.

Allocation	%	GLORY	Release
Play rewards (RewardsPool)	40%	40,000,000	Daily budget with yearly halving
Pool liquidity	15%	15,000,000	95,000 seeded; liquidity reserve
Ecosystem / Team1	15%	15,000,000	Treasury: tournaments, chapters, hackathons, grants
Team (TeamVesting)	15%	15,000,000	1-year cliff, 3-year linear
Community	10%	10,000,000	Airdrops, quests, testers
Reserve	5%	5,000,000	Multisig
Total	**100%**	**100,000,000**	

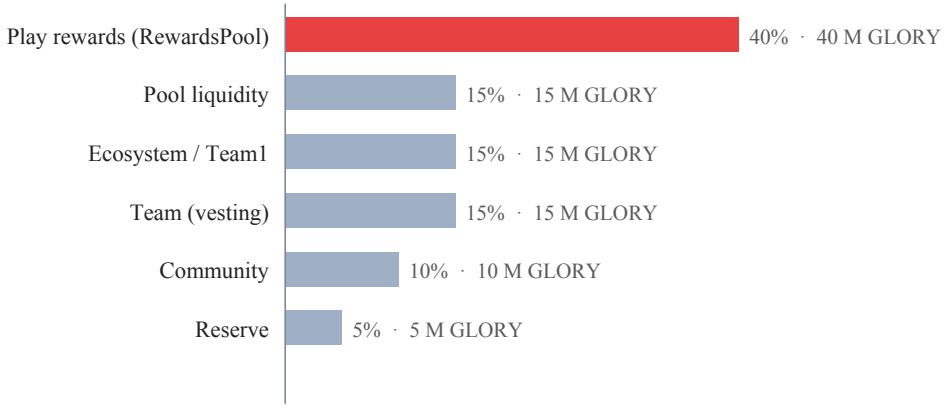


Figure 6. Allocation of the fixed GLORY supply.

6.3 Reward issuance

The RewardsPool never mints: it transfers from its 40 M. The global budget on day d halves every 365 days:

$$B(d) = B_0 \cdot 2^{-a} \tag{2}$$

where a = floor(($d - d_0$) / 365) is the number of full years since deployment and B_0 = 54,794 GLORY.

The reward for a player's n-th victory of the day is:

$$R = min(b \cdot \varphi(n) , C - E_j , B(d) - S_d , balance) \tag{3}$$

where b is the base (500 versus AI, 1,000 in 1 vs 1); $\varphi(n)$ is 1 for victories 1–3, 0.5 for 4–6, 0.25 for 7–10 and 0 afterwards; C = 2,000 is the per-player daily cap, E_j what the player already earned and S_d what was paid that day to everyone. Total issuance is bounded by B(d) regardless of how many accounts exist.

With a base of 500, the first-year budget covers about 110 fully paid victories per day across all players; once it runs out, victories are still recorded but pay 0 until the next day. If activity grows, the admin can lower the base with *MatchRegistry.setParams* without touching supply or the emission ceiling.

Table 9. RewardsPool emission ceiling.

Year	Daily budget	Annual ceiling	Maximum cumulative
1	54,794	~20.0 M	20.0 M
2	27,397	~10.0 M	30.0 M
3	13,698	~5.0 M	35.0 M
4	6,849	~2.5 M	37.5 M
5	3,424	~1.25 M	38.75 M

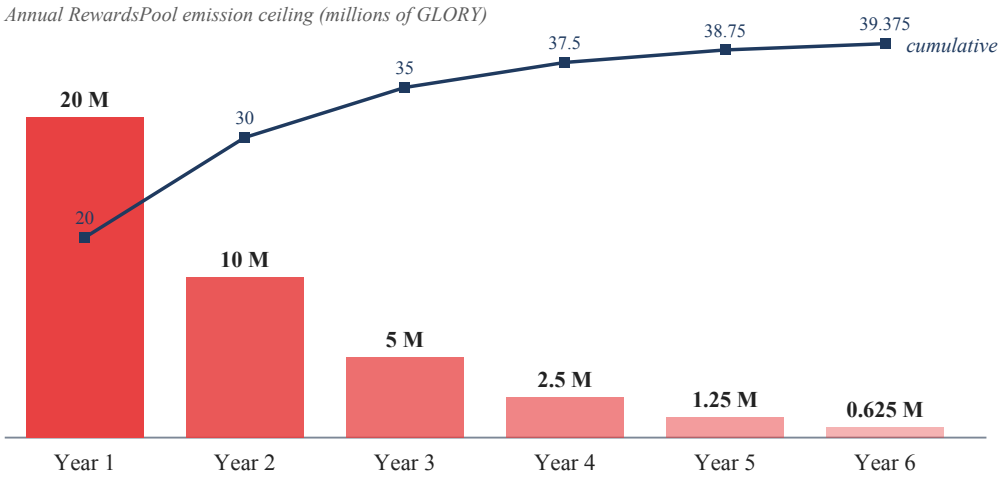


Figure 7. Annual emission ceiling and cumulative. It is a maximum: fewer players means less issuance.

6.4 Sinks

Table 10. GLORY sinks.

Sink	Effect
Official shop (Vault)	50% burned, 50% to the treasury
Army and blueprint NFTs	Burned when brought into a match
Marketplace royalties	5% to the treasury (ERC-2981)
In-match resource purchases	GLORY stays in the pools
Tournaments (Phase 2)	House fee; part may be burned

6.5 Other tokens

Table 11. System tokens.

Token	Type	Supply	Role
AGE	L1 native	Mintable by admin (NativeMinter)	Gas only; sponsored by the protocol. Its role in validator staking will be defined before mainnet
FOOD · WOOD · GOLD · STONE	ERC-20	Elastic, capped	Tokenized resources; minted on settlement, burned on import (see chapter 8)
GameItems	ERC-1155	Max per item	Skins, civilizations, army and blueprints

Chapter 7

Anti-farming and economic security

7.1 On-chain defenses

Table 12. Hard limits in contracts.

Layer	Contract	Stops
Global daily budget with halving	RewardsPool	Mass fake accounts
2,000 GLORY per player per day	RewardsPool	A bot playing all day
Diminishing returns	RewardsPool	Grinding dozens of matches
1 reward per pair per day	MatchRegistry	Collusion between own accounts
No self-matches	MatchRegistry	Winner = loser
Global daily resource cap (50,000)	MatchVault	Draining pools through settlements
500 leftover and 5 settlements per day	MatchVault	Hoarding to dump

7.2 Server-side defenses

Table 13. Verifier filters.

Layer	Stops
Single-use seeds (30 per IP per hour)	Recycled replays and precomputed seeds
Real-time floor	Bots simulating 20 minutes in 2 seconds
Minimum duration (5 min AI, 4 min 1 vs 1)	Express victories
Full re-simulation	Tampered replays
Leftovers only on rewarded victories	Losing or resigning to cash out resources
Easy pays no reward	Farming wins against a weak AI
Guest victories valid for 24 h	Hoarding for mass claims
Wallet signature in rooms	Playing on behalf of another address
Per-IP limits	Abuse of gas, faucet and RPC

7.3 Worst case

Even if an attacker passes every server filter, daily issuance cannot exceed B(d), and settlements cannot extract more than the daily resource cap. Maximum damage is bounded and observable on-chain, and parameters can be tuned by the admin (a multisig on mainnet).

Every defense is tested: 19 contract tests (fixed supply, diminishing returns, caps, halving, collusion and vault cap) plus end-to-end tests of the server and of 1 vs 1.

Chapter 8

Assets, marketplace and pools

8.1 ERC-1155 assets

Each item type is defined on-chain with category, rarity, maximum supply and price. The contract prevents minting above the maximum, even by the team.

Table 14. Rarities and reference prices.

Rarity	Supply per item	Reference price
Common	Unlimited	250–600 GLORY
Rare	1,000–5,000	800–3,000 GLORY
Epic	500	2,500–4,500 GLORY
Legendary	50–100	8,000–9,000 GLORY

Prices were multiplied by 10 in v1.4, when the base reward went from 10 to 500 GLORY: an item costs the same number of victories as before (a legendary, about 16 victories). They are changed with *defineItem* without redeploying.

8.2 Marketplace

Escrowed peer-to-peer trading with partial fills and 5% ERC-2981 royalties to the treasury, paid automatically in the same transaction.

8.3 Resource pools

Table 15. x·y=k pools against GLORY (0.3% fee).

Pool	Initial liquidity	Initial price
FOOD / GLORY	400,000 / 20,000	0.05
WOOD / GLORY	400,000 / 20,000	0.05
GOLD / GLORY	200,000 / 30,000	0.15
STONE / GLORY	250,000 / 25,000	0.10

Selling resources never mints GLORY: it only withdraws it from the pool. Liquidity is protocol-owned, so every swap fee stays in pools the protocol owns.

8.4 Tokenized resources: FOOD, WOOD, GOLD and STONE

The four in-game resources also exist as ERC-20 tokens (*ResourceToken*). They have no fixed supply: only the match vault (*MatchVault*) can mint them, and only with the verifier's signature after re-simulating the match. They are burned when a player buys resources to use inside a match. Outside a match they behave like any ERC-20: they sit in the wallet, show up in the Vault's Resources tab and can be sold or bought in the pools against GLORY.

Table 16. Life cycle of a resource token.

Moment	Contract	What happens
Buy in the in-game Market	MatchVault.buyResource	Pays GLORY to the pool, receives the token and burns it: the resource enters the match
Sell in the in-game Market	MatchVault.sellResource	The resource leaves the match and the sale is recorded; it pays out at settlement
Settle at the end	MatchVault.settle	The verifier signs the amounts; they are minted and, if the player wants, sold in the pool in the same transaction
Keep as tokens	MatchVault.settle	Minted to the player's wallet; they can be sold later in the Vault
Sell or buy outside a match	ResourceMarket	Swap against GLORY (x·y=k, 0.3% fee)

What settles at the end. Whatever was sold in the Market during the match always settles, because the player already paid for it with in-match resources. **Leftovers** (what remains in the stockpile, up to 500 per resource) only settle on a **victory that earns a reward**: a server-issued seed, Normal difficulty or higher and at least 5 minutes of play. On a defeat, a resignation or on Easy, leftovers are lost. 1 vs 1 does not use the on-chain Market, so it has no settlement. Losing therefore never pays more than winning, and resigning cannot be used to cash out resources.

Table 17. Settlement by result.

Result	Sold in match	Leftovers (max 500 each)	Base reward
Victory versus AI (Normal or above)	Yes	Yes	500 GLORY
1 vs 1 victory	N/A	N/A	1,000 GLORY
Victory on Easy or offline	Yes	No	—
Defeat or resignation	Yes	No	—

At the initial pool prices, 500 of each resource is worth about 175 GLORY. Selling resources never mints GLORY: it withdraws it from the pools, so selling pressure is bounded by the leftover cap, 5 settlements per player per day and the global cap of 50,000 units per resource per day. If the pools lose GLORY, resource prices fall on their own (x·y=k) and the treasury can top up liquidity from the 15% reserve.

Chapter 9

Business model

9.1 Principle

Real revenue comes in **AVAX, USDC or card**, not GLORY. GLORY is the currency that circulates inside the game. Tokens are not sold as investments and the game is not funded by the arrival of new players.

9.2 Revenue streams

Table 18. Revenue streams in order of activation.

#	Stream	How it works
1	Ecosystem grants	Avalanche and Team1 programs for L1s and gaming; non-dilutive
2	Sponsored tournaments	Brands and protocols fund the prize pool and appear with their own maps, civilizations or skins
3	Season pass	USD 5–10 per season: cosmetics, missions and chapter houses; no power advantages
4	Cosmetics and primary NFTs	Limited editions in AVAX/USDC; 5% royalties on resales
5	Paid-entry tournaments	USDC or AVAX escrow; 10–15% house fee
6	Protocol-owned liquidity	0.3% fee in protocol pools
7	Licensing and education	"RTS + own L1" template for communities; university editions

9.3 Costs and scalability

Table 19. Cost structure.

Cost	Why it is low / how it scales
Game servers	Matches run in the browser; the server relays commands and re-simulates at the end
Gas	Our own AGE, minted by the protocol
Website	Static (Vercel)
Validators	~1.33 AVAX per month each (ACP-77)
Growth	Split rooms and verifier; re-simulation queue with several workers; more RPC nodes; batched reports

9.4 Illustrative first-year scenario

Table 20. Scenario with explicit assumptions (not a projection).

Stream	Assumption	Year 1
Season pass	10,000 monthly active players, 3% buy, USD 8, 4 seasons	~USD 9,600
Sponsored tournaments	1 per month, USD 3,000 margin	~USD 36,000
Cosmetics and NFTs	Seasonal drops	~USD 5,000–15,000
Grants	One or two programs	Variable

Early on, B2B revenue (sponsorships and grants) outweighs direct sales to players, whose typical free-to-play conversion is 2–5%.

Chapter 10

Roadmap and coming soon

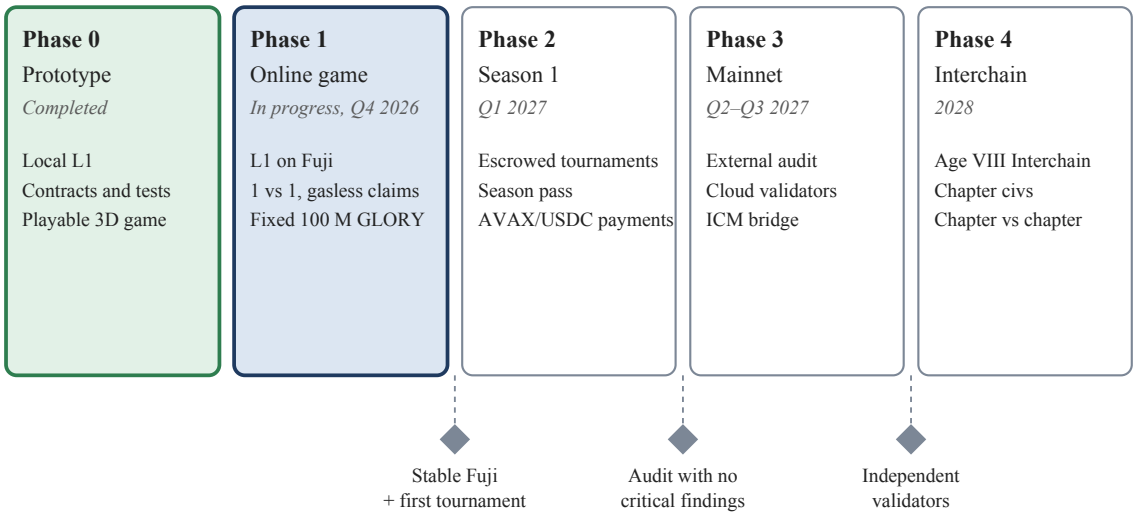


Figure 8. Phased roadmap with its gates.

10.1 Upcoming buildings

- **Conversion and relics:** the Snowman Sage converts enemy units; NFT relics that generate gold.
- **ICM Tower:** Teleporter portals between two owned towers.
- **Validator Data Center:** passive gold in the Futuristic Age.

10.2 Upcoming ages and civilizations

- **VII · Futuristic Age:** already live, with robots, helicopters and plasma tanks.
- **VIII · Interchain Age:** maps connected through ICM, drones and hackers that sabotage buildings, and community battles.
- **One civilization per Team1 chapter** (LatAm, Brazil, India, Korea, Africa…) with their own names, colors and houses.

Chapter 11

Risks and regulatory considerations

Table 21. Risks and mitigations.

Risk	Mitigation
Contract vulnerabilities	Automated tests, external audit before mainnet, OpenZeppelin, multisig
Farming and bots	Layered defenses (Chapter 7); bounded global budget
Compromised verifier key	Dedicated key, KMS in production, dispute window, arbiter, role rotation
Initial centralization	One validator and one server at first; plan for independent validators and servers
Availability (node and server on a single machine)	Self-restarting services (systemd) and a fixed public URL; move to a VPS or the cloud without code changes
Regulatory classification of GLORY	In-game currency not sold as an investment; legal advice per jurisdiction
Prize tournaments	Skill competitions; paid entry only where legal; KYC where applicable

Chapter 12

Conclusions

Age of Avax shows that an RTS can offer real ownership and verifiable rewards without sacrificing the real-time experience: only what needs trust goes on-chain, and deterministic simulation makes any recorded result reproducible and auditable.

Version 1.2 replaces the 1.1 tokenomics with a fixed-supply design, issuance bounded by construction and layered anti-farming defenses, and separates the business (revenue in AVAX, USDC or card) from the game currency. Version 1.3 adds the Futuristic Age, four AI difficulty levels (Easy, with no reward), email sign-in and the ageofavax.app domain. Version 1.4 raises the base reward to 500 GLORY (1,000 in 1 vs 1) with Vault prices ×10, documents the resource tokens and only settles leftovers on victories, so losing never pays more than winning. The immediate work is stable operation on Fuji, escrowed tournaments and an external audit.

References

[1] Ava Labs. *Avalanche L1s*. Avalanche Builder Hub. https://build.avax.network/docs

[2] Avalanche Community Proposals. *ACP-77: Reinventing Subnets*. https://github.com/avalanche-foundation/ACPs

[3] Ava Labs. *Subnet-EVM*. https://github.com/ava-labs/subnet-evm

[4] Radomski, W., et al. (2018). *EIP-1155: Multi Token Standard*.

[5] Burks, Z., et al. (2020). *EIP-2981: NFT Royalty Standard*.

[6] Bloemen, R., Logvinov, L., Evans, J. (2017). *EIP-712: Typed structured data hashing and signing*.

[7] OpenZeppelin. *Contracts 5.x* (ERC20Capped, VestingWalletCliff, AccessControl).

[8] Bettner, P., Terrano, M. (2001). *1500 Archers on a 28.8: Network Programming in Age of Empires and Beyond*. GDC.

Appendix A. Contracts

Table 22. Age of Avax contracts.

Contract	Standard	Role
GloryToken	ERC-20 + Permit + Capped	GLORY, fixed 100 M
RewardsPool	AccessControl	Rewards with budget, caps and halving
TeamVesting	VestingWalletCliff	Team 15 M, 1-year cliff + 3 years
MatchRegistry	EIP-712	Results, disputes, record, claims
MatchVault	EIP-712	Sessions, purchases, sales and settlement
ResourceMarket	AMM x·y=k	4 pools against GLORY
ResourceToken ×4	ERC-20	FOOD, WOOD, GOLD, STONE
GameItems	ERC-1155 + ERC-2981	Catalog, shop with burn, royalties
Marketplace	ERC-1155 escrow	Peer-to-peer trading

Addresses are regenerated on each deployment and published in the repository (game/src/chain/deployment.json).